



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I - III Semester(2025-26)
Faculty Name	Dr. Jayaram Boga
Student Name	Ramannagari Bhanu Priya
Roll Number	2520090171
Branch	CSIT
Course Outcome	6

Case Study Topic: CampusFlow – University Campus Resource Management System using Activity Selection Algorithm

Work Done Summary:

Implemented the Activity Selection Algorithm in Java for efficient campus scheduling and resource management. Modeled campus activities such as classroom sessions, laboratory bookings, seminar hall reservations, and faculty meetings as activities with start and finish times. Applied the greedy Activity Selection Algorithm to select the maximum number of non-overlapping activities and optimize campus resource utilization. Implemented activity insertion, sorting based on finish times, and activity selection operations to manage campus schedules efficiently. Demonstrated how classrooms and facilities can be allocated effectively by selecting compatible activities and avoiding scheduling conflicts. The Activity Selection Algorithm provides efficient performance for campus scheduling, facility booking, and infrastructure optimization in large-scale university management systems.

Implementation Details :

1. Created an Activity class containing activity ID, activity name, start time, and finish time.
2. Stored campus activities such as classroom sessions, laboratory bookings, seminar hall events, and faculty meetings in an array.
3. Sorted activities based on their finish times in ascending order.

4. Implemented the greedy Activity Selection Algorithm to choose non-overlapping activities.
5. Selected the maximum number of compatible activities for efficient facility utilization.
6. Displayed the list of selected campus activities after algorithm execution.
7. Demonstrated conflict-free scheduling and optimal resource allocation using activity selection techniques.
8. The main method inserted campus activity records and executed the Activity Selection Algorithm for campus scheduling.

Result : Screenshots/Output:

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CampusFlow - University Campus Resource Management System
Using Activity Selection Algorithm

Campus Activity Requests:

Activity ID: 101 | Activity: Workshop in Seminar Hall | Time: 9 - 10
Activity ID: 102 | Activity: Placement Training | Time: 9 - 12
Activity ID: 103 | Activity: Coding Contest | Time: 10 - 11
Activity ID: 104 | Activity: Faculty Meeting | Time: 11 - 12
Activity ID: 105 | Activity: Club Event | Time: 12 - 13

Selected Campus Activities:

Activity ID: 101 | Activity: Workshop in Seminar Hall | Time: 9 - 10
Activity ID: 103 | Activity: Coding Contest | Time: 10 - 11
Activity ID: 104 | Activity: Faculty Meeting | Time: 11 - 12
Activity ID: 105 | Activity: Club Event | Time: 12 - 13

Activity Selection Completed Successfully.
```

Time Complexity:

- Sorting Activities: $O(n \log n)$
- Activity Selection Traversal: $O(n)$
- Overall Time Complexity: $O(n \log n)$
- The Activity Selection Algorithm efficiently schedules campus activities and maximizes resource utilization while avoiding booking conflicts.

GitHub Link : <https://github.com/ramannagaribhanupriya/DSA-2-CaseStudies>

Student Signature	Faculty Signature